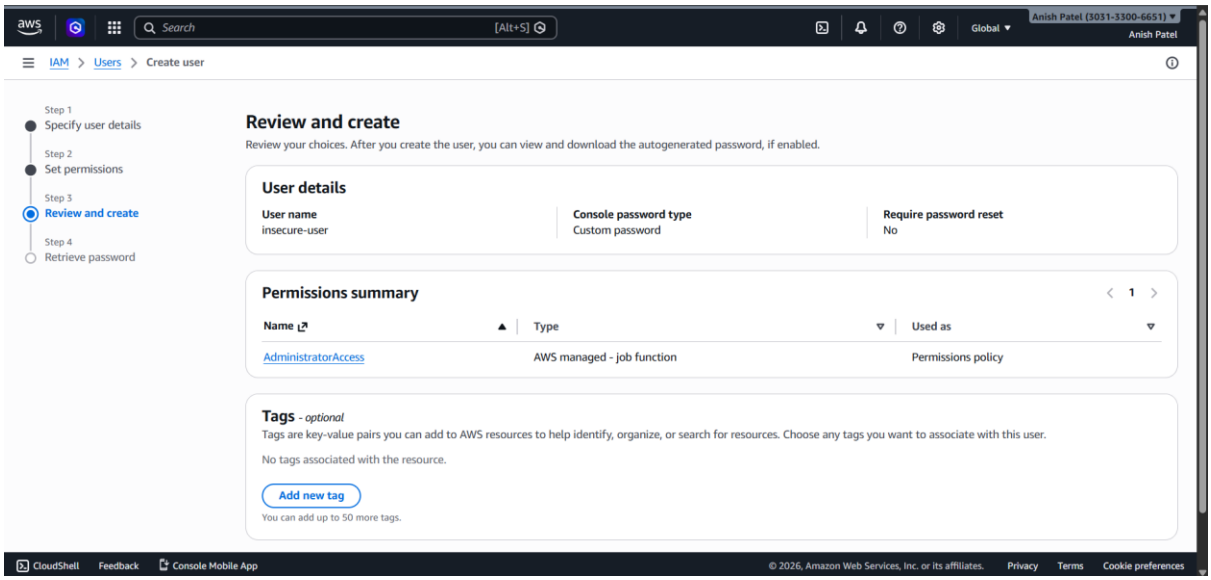
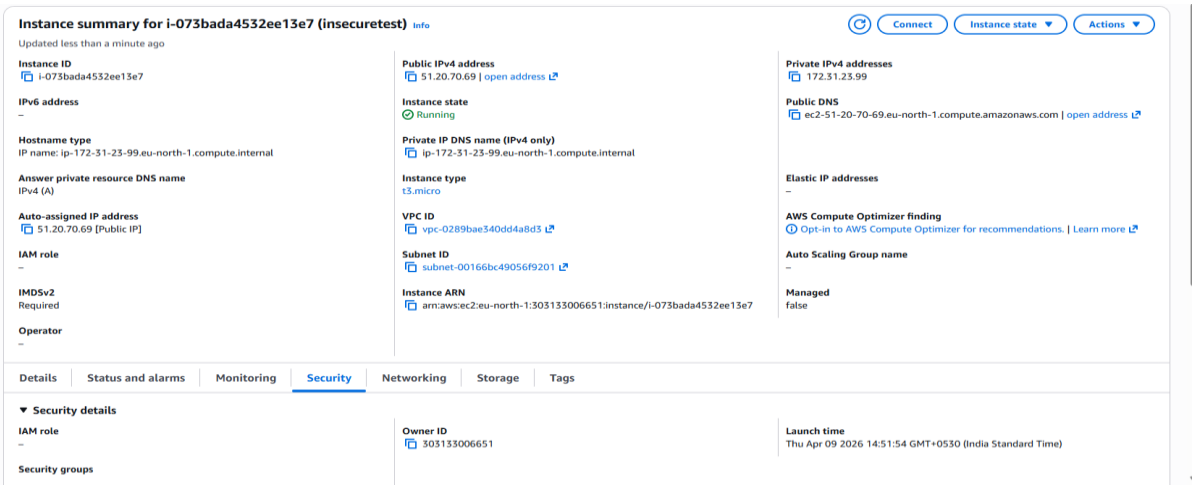
PHASE 1 : Creating Vulnerabilites

Creating user with insecure policies and management:



EC2 instance without any iam role :

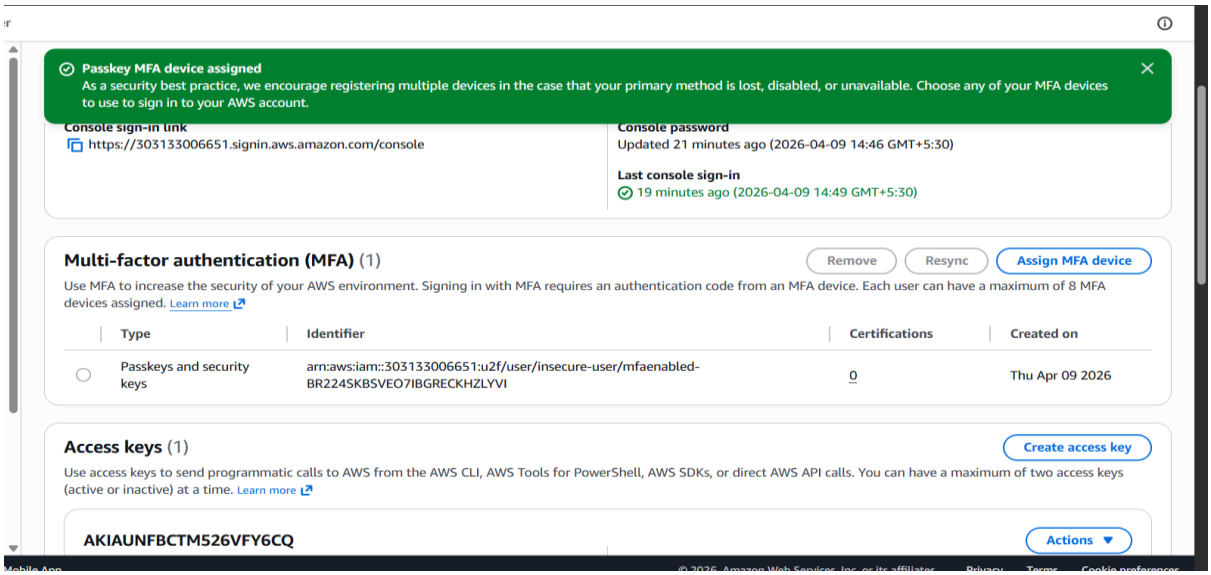


Insecure IAM role showing s3 buckets and iam user credentials:

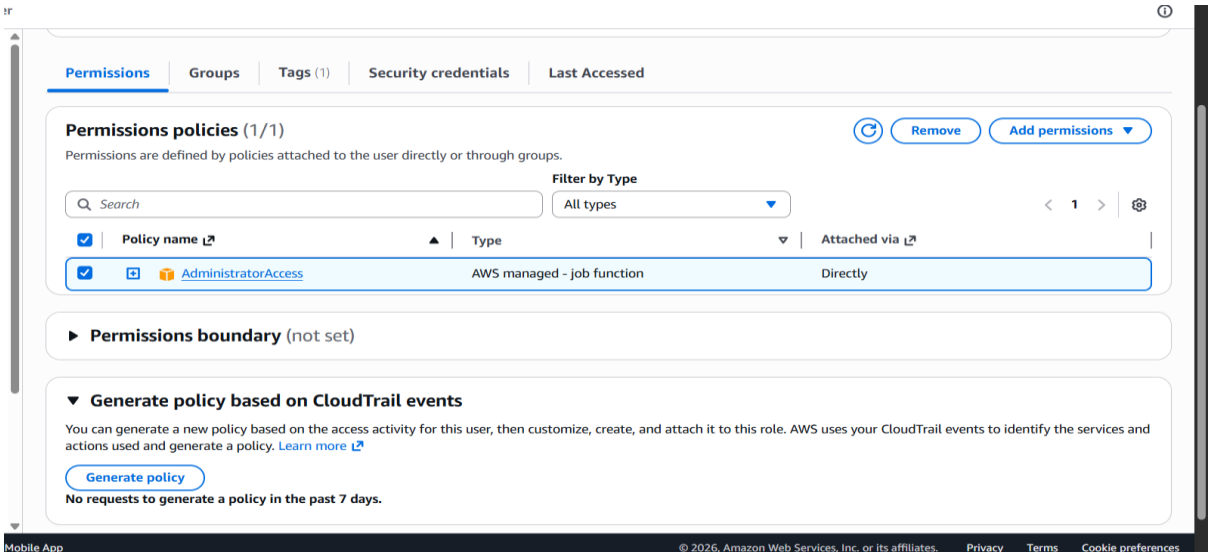
```
AWS Access Key ID [****************Y6CQ]: AWS Secret Access Key [****************TU/q]: Default region name [ap-south-01]: ap-south-1
Default output format [json]: json
[ec2-user@ip-172-31-23-99 ~]$ aws s3 ls
[ec2-user@ip-172-31-23-99 ~]$ aws s3 ls
2026-04-09 09:29:51 insecuretesting
[ec2-user@ip-172-31-23-99 ~]$ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIAUNFBCTM526VFY6CQ
aws_secret_access_key = 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
[ec2-user@ip-172-31-23-99 ~]$
```

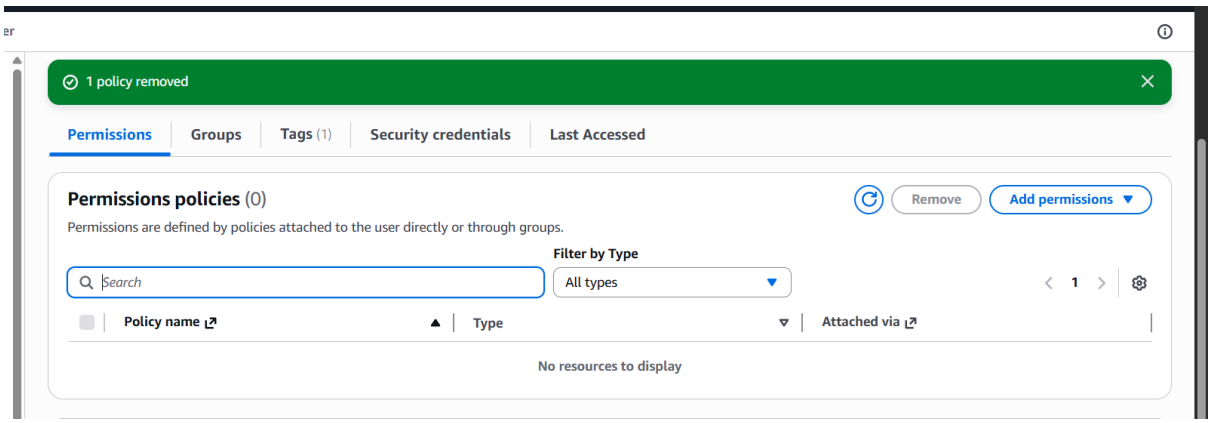
PHASE 2 : Fixing

Enabling MFA on IAM users for High security:

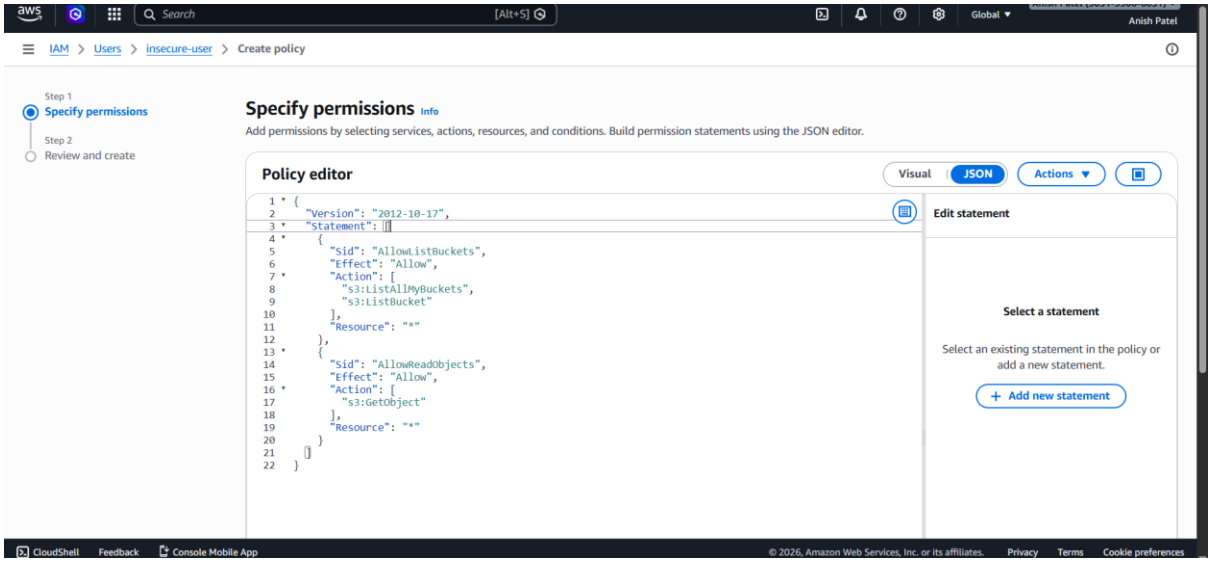


Removing unwanted or over permissible policies:

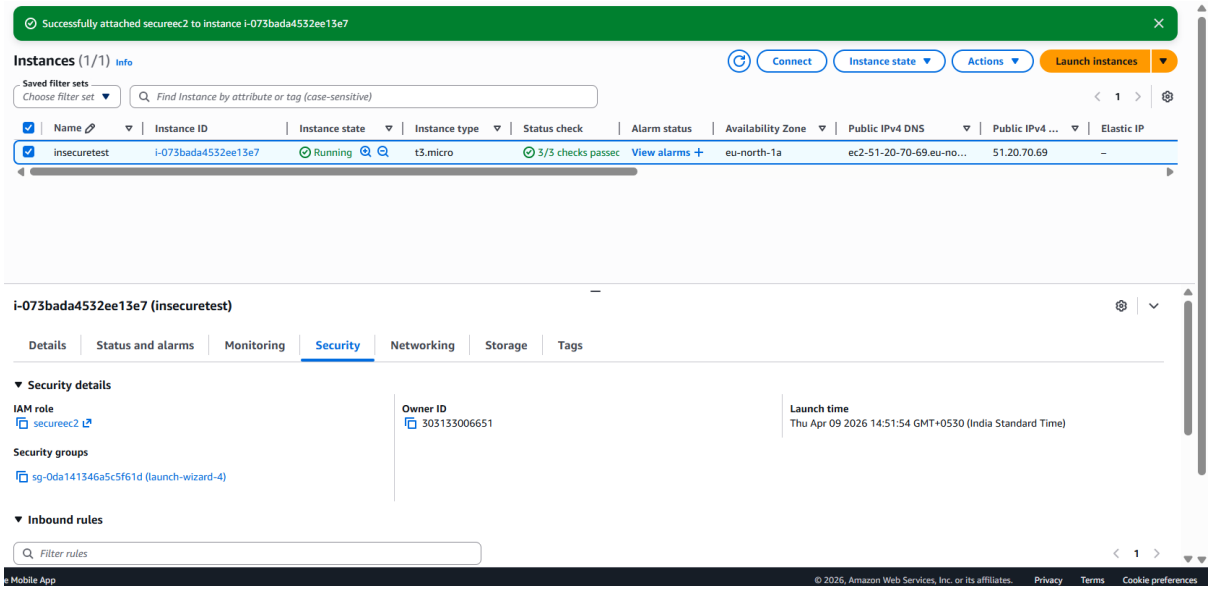




Custom policy for IAM user:



Secured ec2 with iam role:



Final s3 bucket listing without access key proving the iam role working:

aws Search [Alt+S] Ask Amazon Q

```
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
Last login: Thu Apr  9 09:23:04 2026 from 13.48.4.202
[ec2-user@ip-172-31-23-99 ~]$ rm -rf ~/.aws
[ec2-user@ip-172-31-23-99 ~]$ ls
[ec2-user@ip-172-31-23-99 ~]$ cat ~/
cat: /home/ec2-user/: Is a directory
[ec2-user@ip-172-31-23-99 ~]$ aws s3 ls
2026-04-09 09:29:51 insecuretesting
[ec2-user@ip-172-31-23-99 ~]$
```